

SUBLIME docs

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Task 1: Implementing SRHT

I implemented a basic, slightly optimized implementation of the SRHT, following this <https://arxiv.org/pdf/0710.1435>. There is a version of the implementation (of SRHT) similar to the pruned FFT, but according to some website: <https://www.fftw.org/pruned.html>, in the case of the FFT a pruned version may be worse since the un-pruned case is more optimized.

Some results for the chem dataset:

n=13910, d=128, r=1391

R^2 for OLS: 0.998144

R^2 for SRHT: 0.99797

SRHT was consistently faster than OLS but not by much since I've not optimized it fully.

Task 2: Finding how closely fisher information follows the IBOSS upper bound

I keep the mean and stddev predictors and errors the same for all simulations since it doesn't affect anything at all. The value calculated and shown is:

1) Using a normal distribution for the data, predictors and the error (Mean, stddev), rows=1000, cols=51, k=500. Data: (0,2) Predictors:(0,5) Error: (0,1)

Average over 100 values: $9.98664 \cdot 10^{-56}$

2) Using a normal distribution for the data, predictors and the error (Mean, stddev), rows=1000, cols=51, k=1000. Data: (0,2)

Average over 100 values: $1.12938 \cdot 10^{-60}$

3) Using a normal distribution for the data, predictors and the error (Mean,

stddev), rows=1000, cols=51, k=800. Data: (0,2)

Average over 100 values: 9.55571e-59

4) Using a normal distribution for the data, predictors and the error (Mean, stddev), rows=1000, cols=51, k=100. Data: (0,2)

Average over 100 values: 6.75642e-56

5) Using a normal distribution for the data, predictors and the error (Mean, stddev), rows=1000, cols=21, k=500. Data: (0,2)

Average over 100 values: 1.53592e-21

6) Using a normal distribution for the data, predictors and the error (Mean, stddev), rows=1000, cols=11, k=500. Data: (0,2)

Average over 100 values: 5.35963e-09

7) Using a normal distribution for the data, predictors and the error (Mean, stddev), rows=10000, cols=11, k=500. Data: (0,2)

Average over 100 values: 2.75037e-09

8) Using a normal distribution for the data, predictors and the error (Mean, stddev), rows=1000, cols=11, k=500. Data: (0,100)

Average over 100 values: 5.24254e-09

9) Using a uniform distribution for the data, rows=1000, cols=11, k=500. Data: (-2,2)

Average over 100 values: 3.59238e-05

10) Using a uniform distribution for the data, rows=1000, cols=51, k=500. Data: (-2,2)

Average over 100 values: 1.03066e-31

We infer that the bound is not closely followed.

Task 3: Discuss the relation between Fisher Information and R^2

I could not find any literature on this. I performed some numerical simulations instead. The data is chosen from a normal distribution (0,10), the predictors are chosen from a normal distribution (0,2), the error terms are chosen from a normal distribution (0,20). Rows: 10000, Cols: 11, k=100. The average of 100 runs is taken.

log(original fisher information): 99.9222

log(fisher information after IBOSS): 56.5151

R^2 before IBOSS: 0.882303

R^2 after IBOSS: 0.875682

There is clearly a relation between fisher information and R^2 . The reduction in fisher information is clearly much more than the reduction in R^2 . It is difficult to infer what this implies